SHOPSPHERE TECHNOLOGIES PVT. LTD.

Business Requirement Document
Inventory Management Module

ShopSphere E-Commerce Platform — ShopSphere Platform Modernization Program (FY2026)

Document Title	Business Requirement Document — Inventory Management Module
Document ID	BRD-EC-INV-006
Version	1.0
Author	Arjun Rao, Business Analyst — Supply Chain & Inventory
Date	19-Feb-2026
Status	Approved

1. Document Information

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2. Executive Summary

ShopSphere Technologies Pvt. Ltd. operates the ShopSphere E-Commerce Platform, a B2C marketplace serving approximately 2 million monthly active users across India, UAE, and Southeast Asia. As part of the ShopSphere Platform Modernization Program for fiscal year 2026, the Inventory Management Module has been identified as a foundational capability requiring re-architecture to support a single, real-time inventory ledger that acts as the authoritative source of truth for stock availability across all warehouses and fulfillment centers. This document defines the business and functional requirements for the modernized Inventory Management Module, covering real-time stock tracking, stock reservation and release, Available-to-Promise (ATP) calculation, low-stock and out-of-stock alerting, multi-warehouse allocation and rebalancing, and reconciliation against the SAP S/4HANA enterprise resource planning (ERP) ledger. The module is central to preventing overselling during high-demand flash sale events and to ensuring that customers see accurate, trustworthy stock signals on the Product Detail Page (PDP) and Product Listing Page (PLP). This BRD has been reviewed and approved by the Business, Product, IT/Architecture, Quality Assurance, DevOps, and Compliance stakeholder groups and serves as the baseline for downstream functional design, technical architecture, and test planning.

3. Business Objective

The primary business objective of the Inventory Management Module is to establish a unified, real-time, and auditable view of inventory across all ShopSphere fulfillment centers so that the platform can reliably promise, reserve, and fulfil customer orders without overselling or underselling stock. Today, stock visibility is fragmented across the internal Inventory Management System (IMS), NetSuite WMS, and the SAP S/4HANA ERP stock ledger, resulting in periodic discrepancies that lead to cancelled orders, customer dissatisfaction, and manual reconciliation effort by the supply chain team. By modernizing this module, ShopSphere aims to: (a) eliminate overselling incidents during peak sale events such as Diwali and Black Friday when traffic surges up to 10x normal levels; (b) reduce manual stock reconciliation effort between IMS and ERP by at least 70%; (c) improve stock accuracy visible to customers to above 99%; and (d) enable warehouse-level allocation logic that optimizes fulfillment cost and delivery speed across SwiftShip Logistics and Continental Freight Partners. These objectives directly support the program's broader goal of increasing Gross Merchandise Value (GMV) while protecting customer trust and operational margins.

4. Project Scope

In Scope

- Real-time stock tracking across multiple warehouses and fulfillment centers, updated as stock movements occur.
- Stock reservation logic triggered upon order placement, and corresponding stock release logic triggered upon order cancellation, payment failure, or reservation timeout.
- Low-stock and out-of-stock alerting for warehouse operations and merchandising teams, configurable by SKU and warehouse.
- Available-to-Promise (ATP) calculation exposed to the storefront (PDP/PLP) and mobile app to determine sellable quantity per SKU.
- Stock reconciliation workflows against the SAP S/4HANA ERP stock ledger, including discrepancy detection and resolution.
- Multi-warehouse allocation and rebalancing logic to determine the optimal fulfillment source per order and to redistribute stock between warehouses when imbalances are detected.

Out of Scope

- Procurement and purchase order generation to suppliers.
- Supplier contract negotiation and vendor management processes.
- Physical warehouse automation, robotics, and conveyor/pick-pack hardware integration.
- Demand forecasting and AI-driven replenishment recommendation engines.

5. Stakeholders

Name	Role	Department	Responsibility
Rajesh Mehta	VP of E-Commerce	Business	Executive Sponsor
Ananya Sharma	Product Owner, ShopSphere Platform	Product	Prioritization & sign-off
Arjun Rao	Business Analyst, Supply Chain & Inventory	Business Analysis	Requirements owner
Sarah Chen	Enterprise IT Architect	IT/Architecture	Technical feasibility
Michael Torres	QA Lead	Quality Assurance	Test strategy & sign-off
David Kim	DevOps Lead	Engineering/Infra	Deployment & monitoring
Fatima Al-Sayed	Compliance & Security Officer	Risk & Compliance	Regulatory compliance

6. Business Requirements

ID	Business Requirement
BR-001	ShopSphere must maintain a single, real-time inventory ledger that acts as the authoritative source of truth for stock quantities across all warehouses and fulfillment centers.
BR-002	The platform must prevent overselling of any SKU, including during flash sale and peak traffic events, by enforcing accurate stock reservation at the moment of order placement.
BR-003	The business requires automated, near-real-time stock reconciliation between the internal Inventory Management System (IMS) and the SAP S/4HANA ERP stock ledger to eliminate manual reconciliation effort.
BR-004	Warehouse operations and merchandising teams need timely low-stock and out-of-stock alerts to trigger replenishment planning and manage customer-facing stock messaging.
BR-005	The storefront must display accurate Available-to-Promise (ATP) quantities so that customers only see and purchase stock that can realistically be fulfilled within the promised delivery window.
BR-006	ShopSphere requires multi-warehouse allocation logic to determine the optimal fulfillment center for each order based on proximity, stock levels, and logistics partner coverage.
BR-007	The business needs the ability to rebalance stock across warehouses proactively to reduce regional stockouts and improve delivery SLAs.
BR-008	All inventory movements and adjustments must be auditable to support finance, compliance, and dispute resolution requirements.

7. Functional Requirements

Stock Tracking

- `FR-001` The system shall maintain on-hand quantity, reserved quantity, and available quantity per SKU per warehouse, updated in real time as stock events occur.
- `FR-002` The system shall ingest stock movement events (receipts, picks, packs, returns, damages, transfers) from NetSuite WMS and update the central inventory ledger within 5 seconds of the source event.
- `FR-003` The system shall support SKU master data synchronization from the Product Catalog system, including new SKU onboarding and SKU deactivation.
- `FR-004` The system shall provide a warehouse-level and aggregate (platform-wide) view of stock quantities via an internal operations dashboard.

Reservation & Release

- `FR-005` The system shall reserve stock for a SKU at the selected fulfillment warehouse at the moment an order is placed, decrementing the ATP quantity immediately.
- `FR-006` The system shall release reserved stock automatically upon order cancellation, payment failure, or if checkout is not completed within a configurable reservation timeout window (default 15 minutes).
- `FR-007` The system shall support partial reservation release for multi-item orders where only a subset of line items is cancelled or returned.
- `FR-008` The system shall expose real-time stock reservation and release events to the Order Management system via API to keep order status and stock status synchronized.

Alerts

- `FR-009` The system shall generate a low-stock alert when on-hand quantity for a SKU at a warehouse falls below a configurable reorder threshold.
- `FR-010` The system shall generate an out-of-stock alert and automatically suppress or flag the SKU as unavailable on the storefront for the affected warehouse's serviceable region.
- `FR-011` The system shall allow warehouse managers and merchandising staff to configure alert thresholds and notification channels (email, dashboard, SMS) per SKU category.

Available-to-Promise (ATP) & Allocation

- `FR-012` The system shall calculate ATP quantity per SKU as on-hand quantity minus reserved quantity minus safety stock buffer, aggregated across warehouses serviceable to the customer's delivery pincode.
- `FR-013` The system shall expose ATP quantity to the PDP and PLP via API with a response time under 2 seconds at the 95th percentile.
- `FR-014` The system shall apply multi-warehouse allocation rules to select the optimal fulfillment warehouse for each order based on stock availability, proximity to delivery address, and logistics partner (SwiftShip Logistics or Continental Freight Partners) coverage.
- `FR-015` The system shall support scheduled and on-demand stock rebalancing recommendations between warehouses to correct regional stock imbalances.

Reconciliation

- `FR-016` The system shall perform a daily automated reconciliation of stock quantities between the IMS ledger and the SAP S/4HANA ERP stock ledger.
- `FR-017` The system shall flag discrepancies exceeding a configurable variance threshold and route them to a discrepancy resolution workflow for warehouse and finance team review.
- `FR-018` The system shall maintain a full audit trail of all reconciliation adjustments, including reason codes and approving user identity.

8. Non-Functional Requirements

Performance

- `NFR-001` ATP and stock availability API calls shall respond within 2 seconds at the 95th percentile under normal load.
- `NFR-002` Stock synchronization latency between WMS, IMS, and the storefront shall not exceed 5 seconds under normal operating conditions.

Security

- `NFR-003` The module shall comply with OWASP Top 10 secure coding practices for all APIs and internal interfaces.
- `NFR-004` All inventory data at rest shall be encrypted using AES-256, and all data in transit shall use TLS 1.2 or higher.
- `NFR-005` Access to warehouse stock adjustment functions shall be restricted using role-based access control (RBAC), with all adjustments attributable to a named user.

Availability

- `NFR-006` Customer-facing stock availability and ATP services shall maintain 99.9% uptime.
- `NFR-007` Internal warehouse-facing inventory services (WMS integration, reconciliation jobs) shall maintain 99.5% uptime.

Scalability

- `NFR-008` The module shall scale to support peak sale events at up to 10x normal storefront traffic and 5x normal warehouse transaction throughput without degradation of reservation accuracy.
- `NFR-009` The stock reservation engine shall handle concurrent reservation requests for the same SKU without race conditions that could cause overselling.

Reliability

- `NFR-010` The inventory ledger shall support a Recovery Point Objective (RPO) of 15 minutes and a Recovery Time Objective (RTO) of 1 hour for critical inventory services.
- `NFR-011` All stock reservation and release operations shall be idempotent to prevent duplicate reservation or release under retry or network failure conditions.

9. User Stories

- `US-001` As a customer, I want to see accurate stock availability on the Product Detail Page, so that I can trust that the item I order will actually be delivered.
- `US-002` As a warehouse manager, I want to receive a low-stock alert when a SKU falls below its reorder threshold, so that I can initiate replenishment before it goes out of stock.
- `US-003` As an order management system, I want to reserve stock automatically when an order is placed, so that the same unit of inventory is not sold to two customers simultaneously.
- `US-004` As a supply chain analyst, I want a daily reconciliation report comparing IMS and ERP stock ledgers, so that I can identify and resolve discrepancies before they impact fulfillment.
- `US-005` As a fulfillment operations lead, I want the system to automatically select the nearest warehouse with sufficient stock for an order, so that delivery times and shipping costs are optimized.
- `US-006` As a merchandising manager, I want out-of-stock SKUs to be automatically suppressed from the PLP in affected regions, so that customers are not shown items they cannot purchase.
- `US-007` As a customer, I want my reserved stock to be released promptly if I abandon checkout, so that other customers can purchase the item without unnecessary delay.
- `US-008` As a compliance officer, I want every stock adjustment to be logged with a reason code and user identity, so that we can demonstrate an audit trail during regulatory review.

10. Acceptance Criteria

AC ID	Related ID	Criteria	Expected Result
AC-001	FR-005	Given a customer places an order for an in-stock SKU, when the order is confirmed, then stock shall be reserved at the fulfillment warehouse within 2 seconds.	Reserved quantity increments; ATP decrements immediately.
AC-002	FR-006	Given an order is cancelled within 15 minutes of placement, when the cancellation is processed, then reserved stock shall be released automatically.	Reserved quantity decrements; ATP restored within 5 seconds.
AC-003	FR-012	Given multiple warehouses hold stock for a SKU, when ATP is calculated for a customer's pincode, then only serviceable warehouses' stock shall be included.	ATP reflects only deliverable stock for that region.

AC-004	FR-009	Given on-hand quantity for a SKU drops below its configured threshold, when the inventory ledger updates, then a low-stock alert shall be generated within 1 minute.	Alert delivered to configured channel (email/dashboard/SMS).
AC-005	FR-016	Given the daily reconciliation job runs, when IMS and ERP quantities differ beyond the variance threshold, then a discrepancy record shall be created.	Discrepancy appears in resolution workflow queue with variance amount.
AC-006	NFR-009	Given two customers attempt to purchase the last unit of a SKU simultaneously, when both submit orders concurrently, then only one reservation shall succeed.	No overselling occurs; second customer receives an out-of-stock response.
AC-007	FR-014	Given an order is placed with two eligible warehouses in stock, when allocation logic runs, then the warehouse minimizing delivery distance and cost shall be selected.	Order is allocated to optimal warehouse per allocation rules.
AC-008	US-006	Given a SKU reaches zero available stock in a region, when the PLP is rendered for that region, then the SKU shall be marked unavailable or hidden per merchandising configuration.	Customers in the affected region cannot add the SKU to cart.
AC-009	FR-018	Given a manual stock adjustment is made during reconciliation, when the adjustment is saved, then an audit log entry shall capture user ID, timestamp, reason code, and quantity delta.	Audit record is retrievable for compliance review.

11. Business Rules

- `RULE-001` Stock reservations must expire and release automatically if checkout is not completed within 15 minutes, unless overridden by an active promotional hold policy.
- `RULE-002` ATP quantity displayed to a customer must never exceed actual on-hand quantity minus reserved quantity minus safety stock buffer for serviceable warehouses.
- `RULE-003` A SKU shall be marked out-of-stock on the storefront for a given region the instant ATP for all serviceable warehouses in that region reaches zero.
- `RULE-004` Reconciliation discrepancies exceeding 2% of on-hand quantity or 50 units (whichever is lower) for any SKU must be escalated to the Supply Chain & Inventory team within 4 business hours.
- `RULE-005` Manual stock adjustments require a reason code and can only be performed by users holding the Warehouse Supervisor or Inventory Analyst role.
- `RULE-006` Stock rebalancing transfers between warehouses must not reduce the source warehouse's ATP below its configured regional safety stock level.
- `RULE-007` COD (Cash on Delivery) orders shall be subject to the same reservation rules as prepaid orders; reservation is not contingent on payment method.

12. Process Flow Description

Primary Flow — Stock Reservation to Fulfillment:

1. Customer adds a SKU to cart; the storefront queries the ATP service to confirm sellable availability for the delivery pincode.
2. Customer proceeds to checkout and confirms the order; the Order Management system sends a reservation request to the Inventory Management Module.
3. The inventory ledger reserves the requested quantity at the allocated warehouse, decrementing ATP in real time and starting a 15-minute reservation timer.
4. Multi-warehouse allocation logic confirms the optimal fulfillment warehouse based on proximity, stock levels, and logistics partner coverage.
5. Payment is processed; upon successful payment confirmation, the reservation is converted to a committed stock deduction.
6. NetSuite WMS receives the pick instruction and updates stock movement events (pick, pack, ship) back to the inventory ledger.
7. On shipment confirmation, on-hand quantity is permanently decremented and the reserved quantity is cleared for that order line.
8. The SAP S/4HANA ERP stock ledger is updated via the nightly (or near-real-time, where enabled) synchronization job to reflect the fulfilled transaction.

Alternate/Exception Flow — Discrepancy Detected During Reconciliation:

1. The scheduled daily reconciliation job compares IMS on-hand quantities against the SAP S/4HANA ERP stock ledger for each SKU-warehouse combination.
2. If a variance is detected beyond the configured threshold (RULE-004), a discrepancy record is automatically created and routed to the discrepancy resolution queue.
3. A Warehouse Supervisor or Inventory Analyst investigates the discrepancy, cross-checking physical cycle count data from NetSuite WMS.
4. The analyst determines the root cause (e.g., unrecorded damage, mis-pick, transfer-in-transit) and applies a corrective adjustment with a mandatory reason code.
5. The adjustment is logged in the audit trail and propagated to both the IMS ledger and the ERP stock ledger to restore consistency.
6. If the discrepancy cannot be resolved within 4 business hours, it is escalated to the Supply Chain & Inventory Business Analyst for further investigation.

13. Assumptions

- NetSuite WMS will continue to serve as the system of record for physical warehouse stock movements and cycle counts.
- SAP S/4HANA will remain the enterprise system of record for financial inventory valuation.
- All warehouses and fulfillment centers will have reliable network connectivity to support near-real-time stock synchronization.
- Product Catalog SKU master data is accurate and synchronized prior to a SKU becoming sellable on the storefront.
- Logistics partners SwiftShip Logistics and Continental Freight Partners will provide serviceable-pincode data required for ATP and allocation calculations.
- Peak sale event traffic projections (up to 10x normal) provided by the Product team are reasonably accurate for capacity planning.

14. Constraints

- The module must integrate with the existing SAP S/4HANA ERP instance without requiring a parallel ERP migration.
- Reservation and release operations must complete within existing checkout latency budgets defined by the Order Management system.
- Warehouse operations staff have limited capacity for retraining, constraining the complexity of new manual reconciliation workflows.
- Regulatory data residency requirements in India, UAE, and Southeast Asia may restrict where inventory transaction data can be stored and processed.
- The program budget allocated for FY2026 limits the scope of real-time ERP synchronization to daily batch cycles for non-critical SKU categories.

15. Dependencies

- Dependency on the Product Catalog module for accurate and timely SKU master data synchronization.
- Dependency on the Order Management module for order lifecycle events (placement, cancellation, payment confirmation) that trigger reservation and release.
- Dependency on NetSuite WMS for physical stock movement events and cycle count data feeding the inventory ledger.
- Dependency on the SAP S/4HANA ERP system for financial stock valuation and the daily reconciliation baseline.
- Dependency on logistics partners SwiftShip Logistics and Continental Freight Partners for serviceable-area data used in warehouse allocation.

16. Risks

Risk ID	Description	Likelihood	Impact	Mitigation
RISK-001	Overselling during flash sale events due to reservation race conditions under 10x peak load.	Medium	High	Implement atomic, distributed locking on stock reservation and load-test at 10x projected peak traffic.

RISK-002	Persistent discrepancies between IMS and ERP ledgers erode trust in stock accuracy reporting.	Medium	High	Automate daily reconciliation with escalation SLAs and root-cause tracking.
RISK-003	Delayed stock synchronization from NetSuite WMS causes stale ATP shown to customers.	Low	Medium	Enforce 5-second sync latency SLA with monitoring and alerting via DevOps.
RISK-004	Incorrect warehouse allocation increases shipping cost and delivery time.	Medium	Medium	Continuously tune allocation rules using logistics partner performance data.
RISK-005	Unauthorized manual stock adjustments could mask shrinkage or fraud.	Low	High	Enforce RBAC, mandatory reason codes, and full audit logging for all adjustments.
RISK-006	Regional data residency regulations delay cross-border inventory data replication.	Low	Medium	Engage Compliance & Security Officer early to design region-aware data storage architecture.

17. Data Requirements

The Inventory Management Module must capture and persist the following core data entities, sourced from and synchronized with the Product Catalog, NetSuite WMS, and SAP S/4HANA systems. Historical stock movement and reservation records must be retained for a minimum of 7 years to satisfy financial audit and compliance requirements, while real-time ledger snapshots are retained indefinitely for reporting purposes.

Field	Description
SKU	Unique stock keeping unit identifier synchronized from Product Catalog.
Warehouse ID	Unique identifier for the warehouse or fulfillment center holding the stock.
On-Hand Quantity	Physical quantity of the SKU currently present at the warehouse.
Reserved Quantity	Quantity currently held against unfulfilled or in-progress orders.
Available to Promise (ATP)	Sellable quantity calculated as on-hand minus reserved minus safety stock.
Safety Stock Buffer	Configurable minimum quantity withheld from ATP to absorb variance.
Reorder Threshold	Quantity level triggering a low-stock alert.
Last Reconciliation Timestamp	Date and time of the last successful IMS-ERP reconciliation for the SKU-warehouse pair.

18. Integration Requirements

System	Purpose	Direction	Data Exchanged
SAP S/4HANA (ERP)	Inventory valuation and daily stock ledger reconciliation	Bi-directional	On-hand quantities, stock valuation, reconciliation adjustments
NetSuite WMS	Physical stock movement tracking and cycle counts	Inbound to IMS	Receipts, picks, packs, transfers, cycle count results
Product Catalog	SKU master data synchronization	Inbound to IMS	SKU attributes, active/inactive status, unit of measure
Order Management	Real-time stock reservation and release triggered by order lifecycle events	Bi-directional	Order ID, SKU, quantity, reservation status, cancellation events

19. Reporting Requirements

- Real-time stock availability dashboard by SKU and warehouse for warehouse operations teams.
- Daily IMS-to-ERP reconciliation summary report highlighting discrepancies and resolution status.
- Low-stock and out-of-stock alert log with response time metrics, published via Tableau.
- Oversell incident report tracking frequency, affected SKUs, and root cause during peak sale events.
- Warehouse allocation and rebalancing effectiveness report, tracking delivery time and cost impact.
- Stock accuracy trend report combining IMS, WMS, and ERP data sources for executive review.

20. Success Metrics

KPI	Target
Stock accuracy (IMS vs. physical count)	≥ 99%
Oversell incidents per month	≤ 2 platform-wide
IMS-to-ERP reconciliation discrepancy resolution time	≤ 4 business hours

ATP API response time (95th percentile)	< 2 seconds
Stock synchronization latency	< 5 seconds
Manual reconciliation effort reduction	≥ 70% reduction vs. current-state baseline

21. Future Enhancements

- AI-driven demand forecasting and automated replenishment recommendations (currently out of scope).
- Predictive stock rebalancing using machine learning to anticipate regional demand shifts ahead of sale events.
- Integration with supplier-managed inventory (VMI) for select high-velocity SKU categories.
- Real-time (sub-minute) ERP synchronization for all SKU categories, expanding beyond the current daily batch cycle.
- Automated dynamic safety stock calculation based on regional demand volatility.

22. Glossary

Term	Definition
SKU	Stock Keeping Unit; a unique identifier for a sellable product variant.
PDP	Product Detail Page; the storefront page displaying detailed information about a single product.
PLP	Product Listing Page; the storefront page displaying a list or grid of products.
ATP	Available to Promise; the quantity of a SKU that can reliably be sold and delivered.
GMV	Gross Merchandise Value; the total value of goods sold through the platform.
COD	Cash on Delivery; a payment method where the customer pays upon receipt of goods.
RMA	Return Merchandise Authorization; the process governing product returns.
SLA	Service Level Agreement; a defined commitment for service performance or response time.
IMS	Inventory Management System; ShopSphere's internal system tracking stock quantities.
WMS	Warehouse Management System; NetSuite WMS, used to manage physical warehouse operations.
Safety Stock	A buffer quantity withheld from ATP to guard against demand or supply variability.

23. Appendix

Reference Documents: ShopSphere Platform Modernization Program Charter (FY2026); ShopSphere Order Management BRD (BRD-EC-OMS-004); ShopSphere Product Catalog BRD (BRD-EC-CAT-002); ShopSphere Logistics & Fulfillment BRD (BRD-EC-LOG-007).

Version	Date	Author	Change Description
0.1	02-Feb-2026	Arjun Rao	Initial draft circulated for stakeholder review.
0.5	11-Feb-2026	Arjun Rao	Incorporated feedback from IT/Architecture and QA on ATP and reconciliation requirements.
1.0	19-Feb-2026	Arjun Rao	Final version approved by Rajesh Mehta and Ananya Sharma.